

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

Department of Electronics & Electrical Communication Engineering



SNAP - FPGA

DESIGN LAB

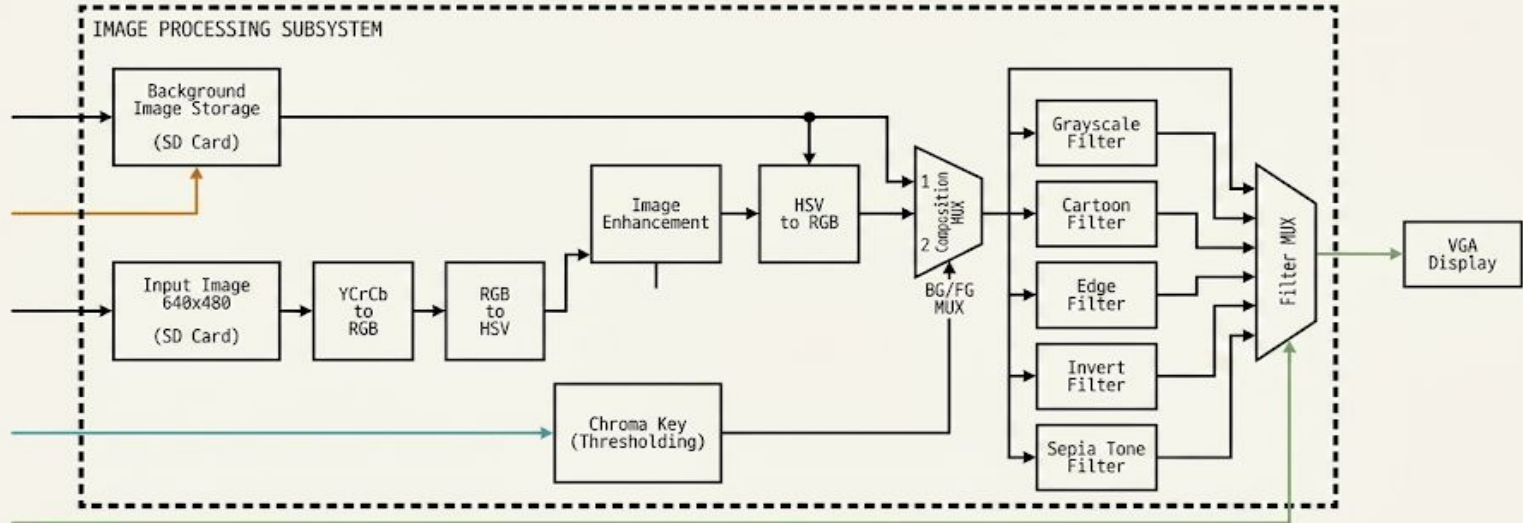
Group Number: 02

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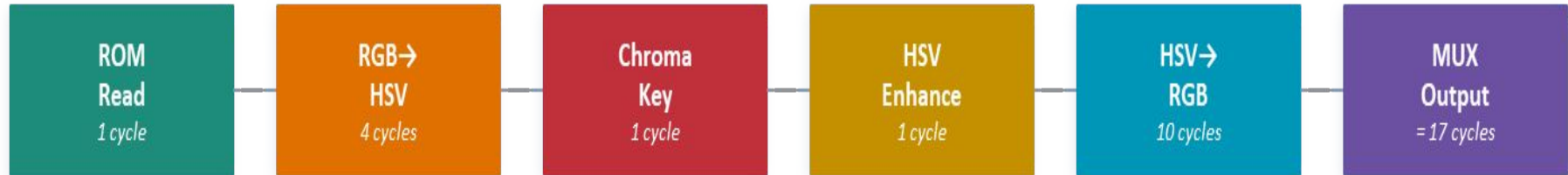
Supervisor:
Prof. Aniket Singha



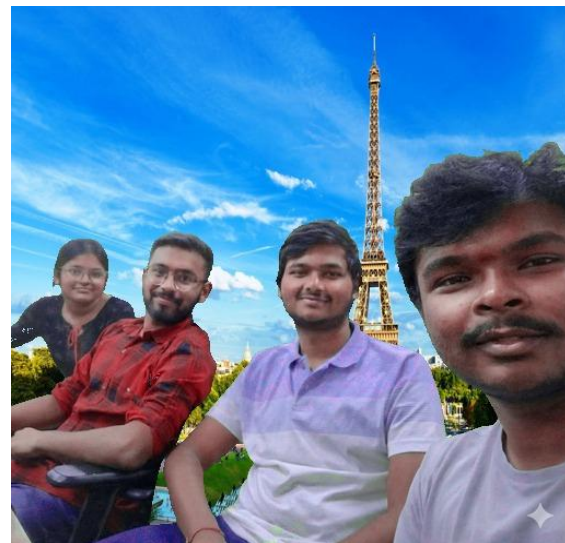
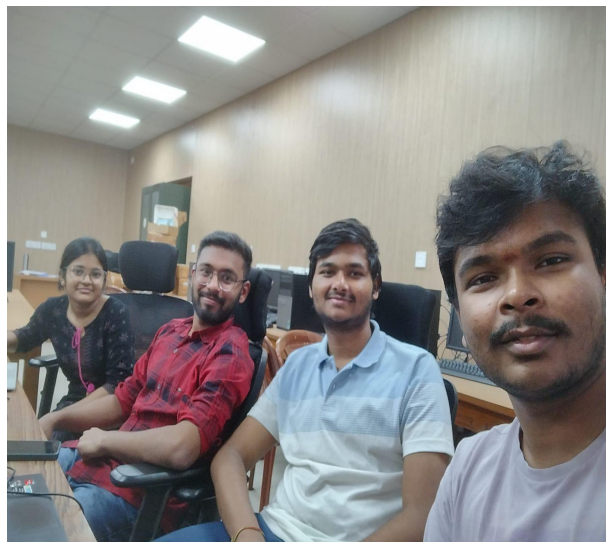
BLOCK DIAGRAM



STAGE - 01: BACKGROUND ADDITION



BACKGROUND ADDITION





STAGE - 02 : FILTERS AND EQUATIONS

SW[0]
filters_en
Enable Stage 2

SW[1]
enhance_en
HSV brightness

SW[3:2]=00
SEPIA
Warm matrix

SW[3:2]=01
INVERT
255-channel

SW[3:2]=10
EDGE
Sobel gradient

SW[3:2]=11
CARTOON
Gauss + Sobel

SEPIA (SW[3:2]=00)

$R_{out} = 0.393R + 0.769G + 0.189B$
 $G_{out} = 0.349R + 0.686G + 0.168B$
 $B_{out} = 0.272R + 0.534G + 0.131B$

INVERT (SW[3:2]=01)

$R' = 255 - R$
 $G' = 255 - G$
 $B' = 255 - B$

EDGE — Sobel (SW[3:2]=10)

$G_x = (a_2 + 2a_3 + a_4) - (a_0 + 2a_7 + a_6)$
 $G_y = (a_0 + 2a_1 + a_2) - (a_6 + 2a_5 + a_4)$
 $grad = |G_x| + |G_y| \quad thr=50 \rightarrow \text{binary}$

CARTOON = Gaussian + Sobel (SW[3:2]=11)

Kernel: $\begin{matrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{matrix} \quad (\div 16)$
Sobel edge > 100 $\rightarrow$ black outline overlay

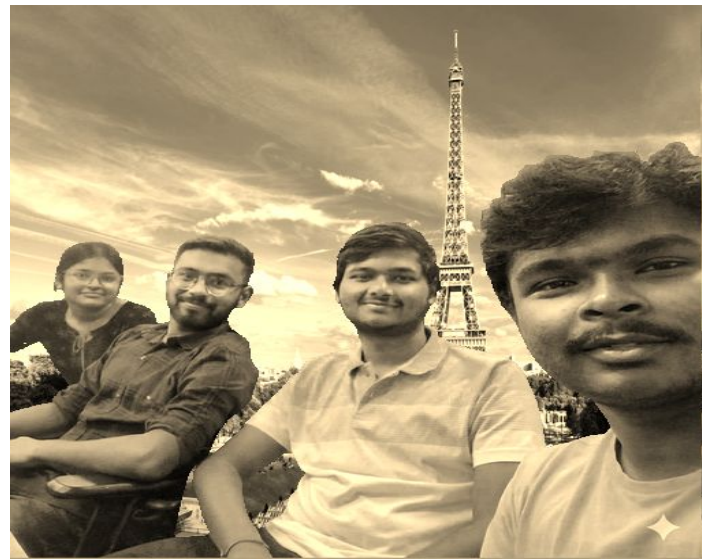
WITHOUT FILTER



GREY SCALE FILTER



SEPIA FILTER



INVERT FILTER



EDGE DETECTION FILTER



CARTOON FILTER

